

NMAM INSTITUTE OF TECHNOLOGY, NITTE

Off-Campus Centre of Nitte (Deemed to be University)

Third Semester B. Tech (CBCS) Degree Examinations

Summer Semester July 2024

AM1104-1 – INTRODUCTION TO MACHINE LEARNING

(AIML)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- What is true about Machine Learning?**
 - Machine Learning (ML) is that field of computer science
 - ML is a type of artificial intelligence that extract patterns out of raw data by using an algorithm or method
 - The main focus of ML is to allow computer systems learn from experience without being explicitly programmed or human intervention.
 - All the above
- Which of the following are ML methods?**
 - Supervised Learning
 - Reinforcement Learning
 - Semi Machine Learning
 - Unsupervised Learning
- Which of the following is a disadvantage of decision trees?**
 - Factor analysis
 - Decision trees are robust to outliers
 - Decision trees are prone to be overfit
 - None of the above
- How do you handle missing or corrupted data in a dataset?**
 - Drop missing rows or columns
 - Replace missing values with mean/median/mode
 - Assign a unique category to missing values
 - All of the above
- ML is a field of AI consisting of learning algorithms that?**
 - Improve their performance
 - At executing some task
 - Over time with experience
 - All of the above
- What is the difference between supervised and unsupervised learning?**
 - Supervised learning requires labeled data while unsupervised learning does not.
 - Unsupervised learning requires labeled data while supervised learning does not.
 - Supervised learning does not require data while unsupervised learning does.
 - There is no difference between supervised and unsupervised learning.
- Which of the following is an example of a classification problem?**
 - Predicting the price of a house based on its features
 - Predicting the weight of a person based on their height
 - Predicting whether a customer will churn or not
 - Predicting the age of a person based on their income
- What is the purpose of feature scaling in machine learning?**
 - To convert categorical features into numerical features
 - To reduce the dimensionality of the feature space
 - To standardize the range of numerical features
 - To introduce new features into the model

9. What is the relation between a candidate and frequent itemset?
 A) A candidate itemset is always a frequent itemset B) A frequent itemset must be a candidate itemset
 C) No relation between these two D) Strong relation with transactions
10. For the question given below consider the data Transactions :
 I1, I2, I3, I4, I5, I6
 I7, I2, I3, I4, I5, I6
 I1, I8, I4, I5
 I1, I9, I10, I4, I6
 I10, I2, I4, I11, I5
 With support as 0.6 find all frequent item sets?
 A) $\langle I1 \rangle, \langle I2 \rangle, \langle I4 \rangle, \langle I5 \rangle, \langle I6 \rangle, \langle I1, I4 \rangle, \langle I2, I4 \rangle, \langle I2, I5 \rangle, \langle I4, I5 \rangle, \langle I4, I6 \rangle, \langle I2, I4, I5 \rangle$ B) $\langle I2 \rangle, \langle I4 \rangle, \langle I5 \rangle, \langle I2, I4 \rangle, \langle I2, I5 \rangle, \langle I4, I5 \rangle, \langle I2, I4, I5 \rangle$
 C) $\langle I11 \rangle, \langle I4 \rangle, \langle I5 \rangle, \langle I6 \rangle, \langle I1, I4 \rangle, \langle I5, I4 \rangle, \langle I11, I5 \rangle, \langle I4, I6 \rangle, \langle I2, I4, I5 \rangle$ D) $\langle I1 \rangle, \langle I4 \rangle, \langle I5 \rangle, \langle I6 \rangle$
11. What will happen if support is reduced?
 A) Number of frequent item sets remains the same B) Some item sets will add to the current set of frequent item sets
 C) Some item sets will become infrequent while others will become frequent D) Can not say
12. Which of the following is finally produced by Hierarchical Clustering?
 A) final estimate of cluster centroids B) tree showing how close things are to each other
 C) assignment of each point to clusters D) all of the mentioned
13. Which of the following is required by K-means clustering
 A) defined distance metric B) number of clusters
 C) initial guess as to cluster centroids D) all of the mentioned
14. Which of the following clustering requires merging approach?
 A) Partitional B) Hierarchical
 C) Naive Bayes D) None of the mentioned
15. In which of the following cases will K-means clustering fail to give good results?
 1) Data points with outliers
 2) Data points with different densities
 3) Data points with nonconvex shapes
 A) 1 and 2 B) 2 and 3
 C) 1 and 3 D) All of the above
16. Which of the following is an example of a clustering algorithm?
 A) Decision tree B) Random forest
 C) K-means D) Gradient descent
17. Which of the following is a common approach to reducing dimensionality in machine learning?
 A) Feature selection B) Feature extraction
 C) Feature scaling D) Feature engineering
18. Which of the following is a reasonable way to select the number of principal components "k"?
 A) Choose k to be the smallest value so that at least 99% of the variance is retained. B) Choose k to be 99% of m ($k = 0.99 * m$, rounded to the nearest integer).
 C) Choose k to be the largest value so that 99% of the variance is retained. D) Use the elbow method.
19. When performing regression or classification, which of the following is the correct way to preprocess the data?
 A) Normalize the data -> PCA -> training B) PCA -> normalize PCA output -> training
 C) Normalize the data -> PCA -> normalize PCA output -> training D) None of the above
20. Which of the following is a widely used and effective machine learning algorithm based on the idea of bagging?
 A) Decision Tree B) Regression
 C) Classification D) Random Forest

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

1. a) Define machine Learning. Explain different types of machine learning.
 b) Discuss the different types of attributes based on measurement scale.
 c) What is Sampling and explain its types.
2. a) Write Hunts algorithm and apply the same to construct decision tree for the following data

	binary	categorical	continuous	class
Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Marks	BT*	CO*	PO*
8	L*2	1	1
4	L2	1	1
4	L2	1	1

- b) With a neat diagram, explain feature subset selection methodology.

8	L2	2	1
8	L2	2	1

3. a) For the dataset, compute information gain for the attributes Humidity and Wind. Comment which among them will be selected for splitting.

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

8	L2	2	1
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- b) Write the algorithm for KNN and predict the following test data based on KNN

Test Data (T): <Acid Durability=4, Strength=8, Class=?>

Name	Acid Durability	Strength	Class
Type 1	4	3	Good
Type 2	6	5	Bad
Type 3	3	7	Bad
Type 4	2	8	Good

8	L3	CO2	2
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Unit – II

9. 4. a) Explain the process of Frequent Itemset Generation using the Apriori Algorithm. 8 L3 3 2
- 10 b) Apply Apriori algorithm to generate frequent itemsets for the following data. Given minimum support is 50%. What is maximum number of association rules that can be extracted from the data?

TID	Items
1	{Milk, Eggs, Cola}
2	{Bread, Butter, Milk}
3	{Milk, Cola, Cookies}
4	{Bread, Butter, Cookies}
5	{Eggs, Cookies, Cola}
6	{Milk, Cola, Bread, Butter}
7	{Bread, Butter, Cola}
8	{Eggs, Cola}
9	{Milk, Cola, Bread, Butter}
10	{Eggs, Cookies}

8 L3 3 2

- 5 a) With an example show the use of naïve bayes classification. 8 L2 3 1
- b) Discuss the Importance of Support vector machine in machine learning. 8 L2 3 1
- 6 a) Explain DBSCAN algorithm with an example 8 L2 4 1
- b) For the given proximity matrix, construct dendrogram and nested cluster representation using single link (MIN) agglomerative hierarchical clustering (AHC) method

	A	B	C	D	E	F
A	0.00	0.71	5.66	3.61	4.24	3.20
B	0.71	0.00	4.95	2.92	3.54	2.50
C	5.66	4.95	0.00	2.24	1.41	2.50
D	3.61	2.92	2.24	0.00	1.00	0.50
E	4.24	3.54	1.41	1.00	0.00	1.12
F	3.20	2.50	2.50	0.50	1.12	0.00

8 L2 4 2

Unit – III

7. a) Show the working of random forest algorithm. Explain the applications of random forest algorithm in brief. 8 L2 5 1
- b) Compare the bagging and boosting. 8 L2 5 1
8. a) Discuss how to reduce dimensionality using factor analysis. 8 L2 5 1
- b) Explain different method of solving imbalance problem. 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
